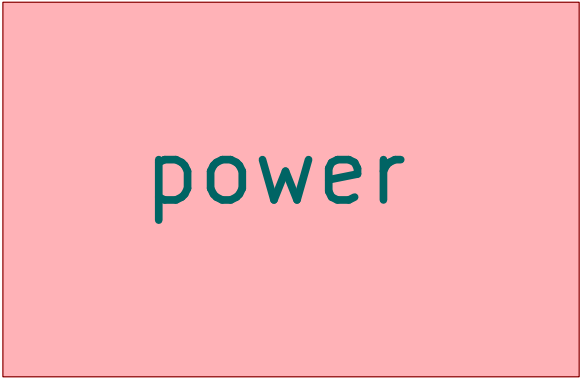




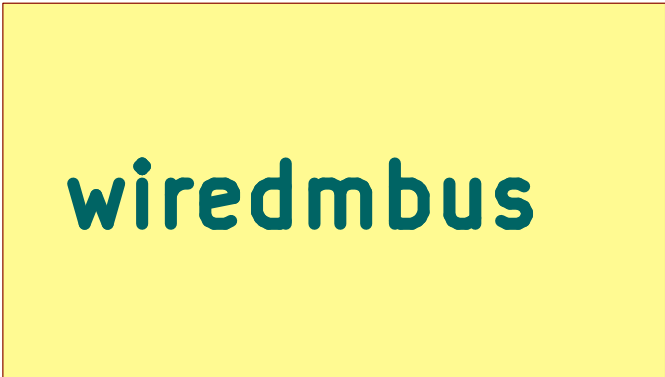
File: power.kicad_sch



File: connector.kicad_sch



File: ethernet.kicad_sch



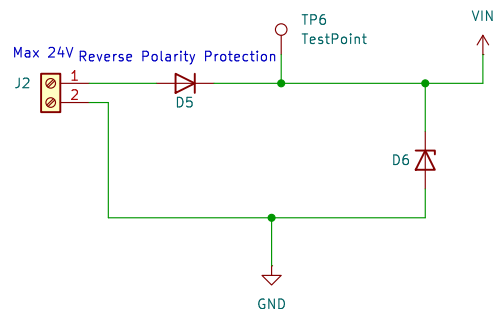
File: mbus.kicad_sch



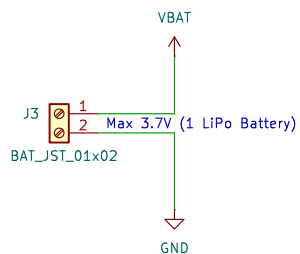
File: sdcard.kicad_sch

Approved By: Rodney Osodo Designed By: Jones Kisaka Abstract Machines		
Sheet: / File: s0-base-board.kicad_sch		
Title: S0 Base Board		
Size: A4	Date: 2025-11-26	Rev: v0.3.0
KiCad E.D.A. 9.0.6		Id: 1/6

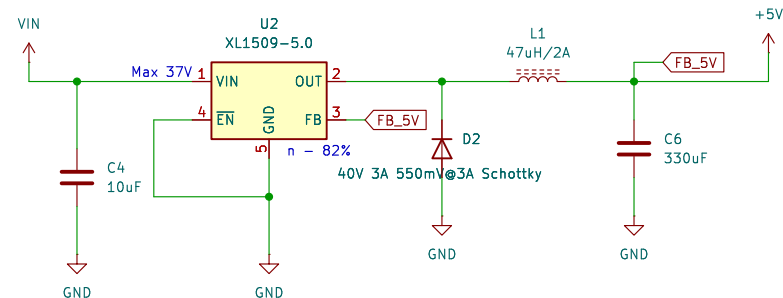
Reverse Polarity Protection



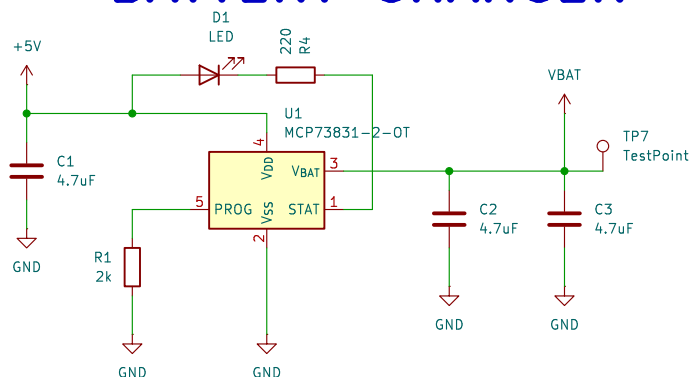
BAT



VIN – 5V CONV



BATTERY CHARGER



CURRENT REGULATION SET

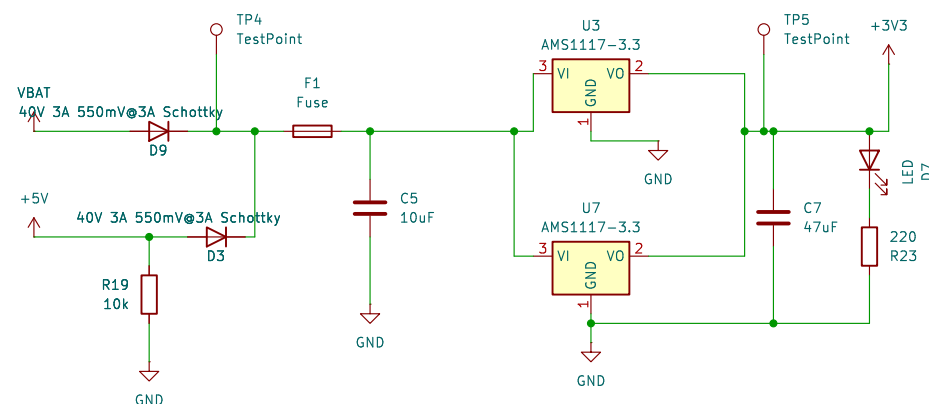
Fast charge current regulation can be scaled by placing a programming resistor (RPROG) from the PROG input to VSS. The program resistor and the charge current are calculated using the following equation:

$$I_{reg} = 1000V/R_{prog}$$

where: R_{prog} is in kOhms
 I_{reg} is charging current in milliamps.

$$I_{reg} = 1000V/2 = 500mA$$

5V – 3.3V CONV



Approved By: Rodney Osodo

Designed By: Jones Kisaka

Abstract Machines

Sheet: /power/

File: power.kicad_sch

Title: S0 Base Board

Size: A4 Date: 2025-11-26

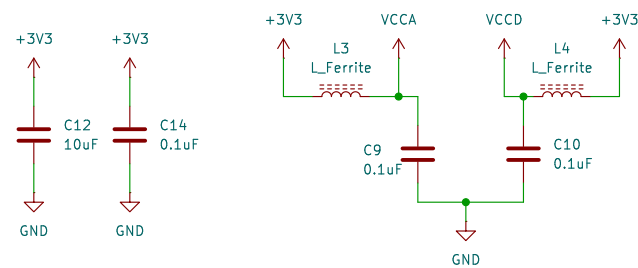
KiCad E.D.A. 9.0.6

Rev: v0.3.0

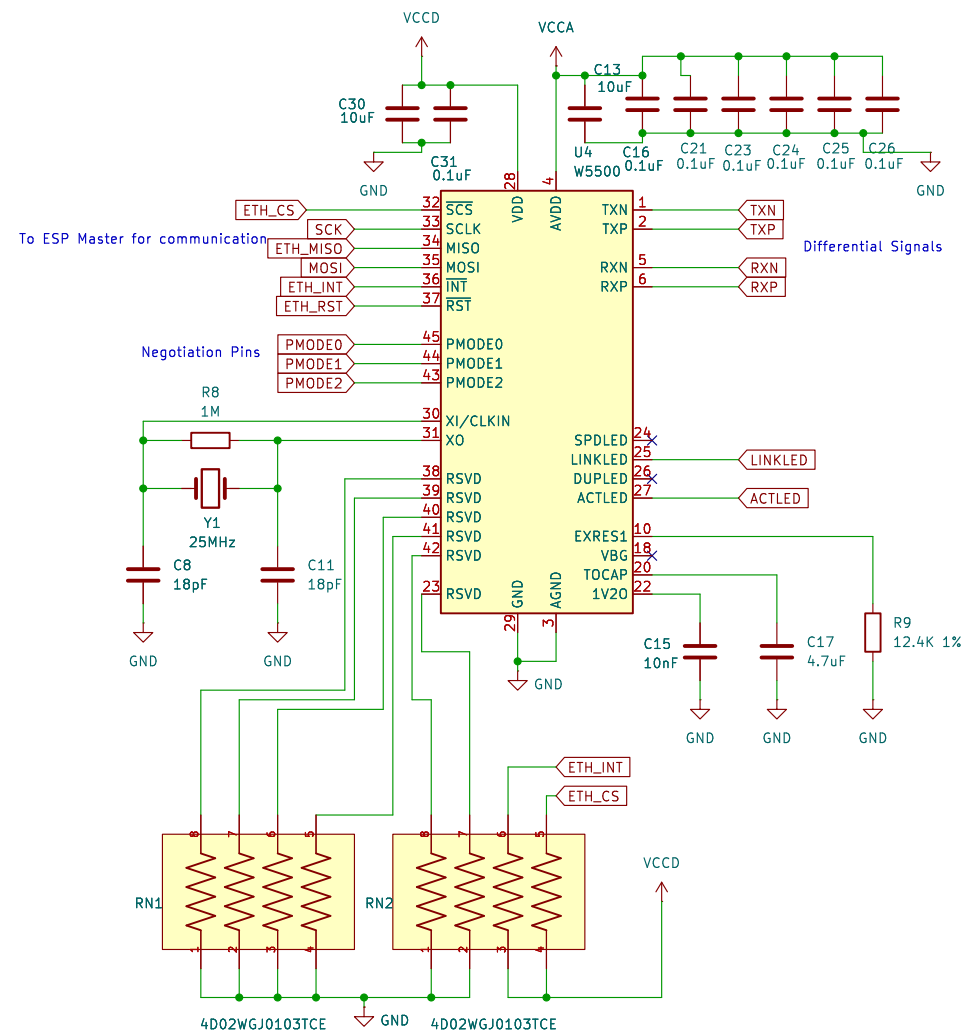
Id: 2/6

DECOUPLING CAPS

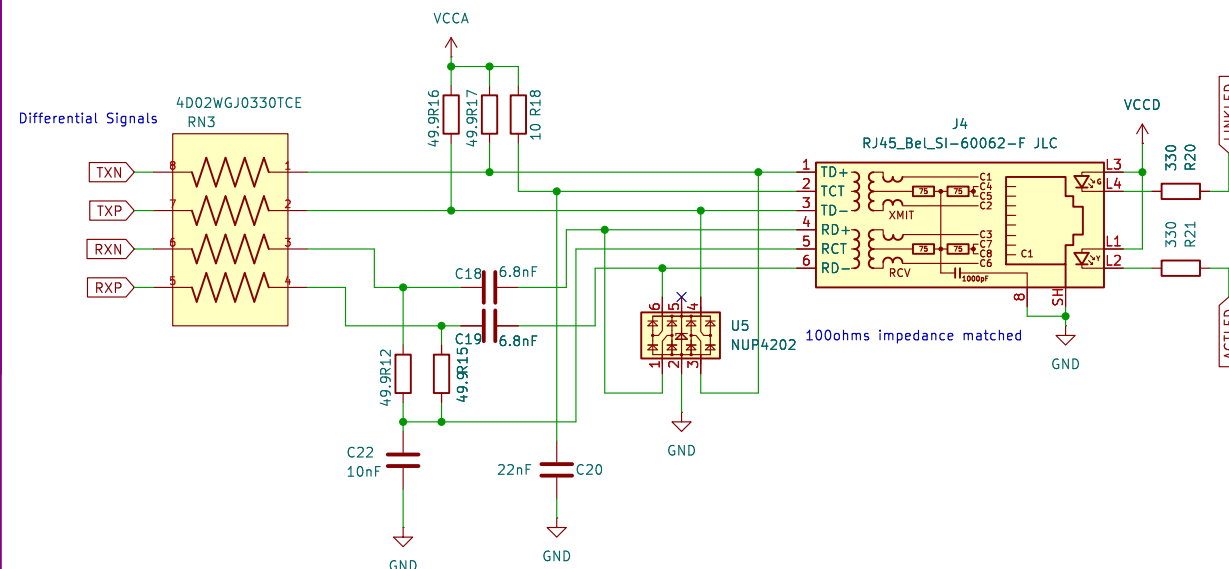
Decoupling Caps help ensure stable operation
Filtering noise
Maintain a steady power supply.



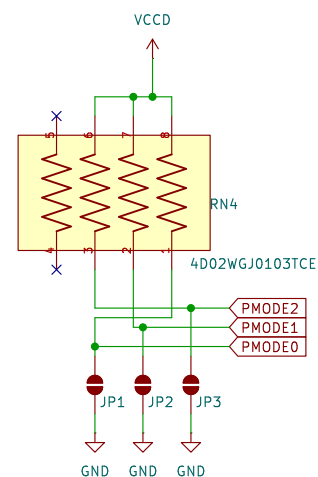
W5500 ETHERNET PHY/MAC



JACK & MAGNETICS



AUTO-NEGOTIATING MODES

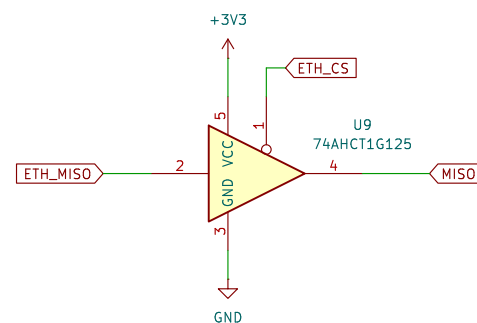


Negotiation of Speed and Connection type ie Full or Half Duplex through these pins.

You can set:

- 10 Mbps Half-Duplex
- 10 Mbps Full-Duplex
- 100 Mbps Half-Duplex
- 100 Mbps Full-Duplex

SPI MISO ISOLATOR



Approved By: Rodney Osodo
Designed By: Jones Kisaka
Abstract Machines
Sheet: /Ethernet/
File: ethernet.kicad_sch

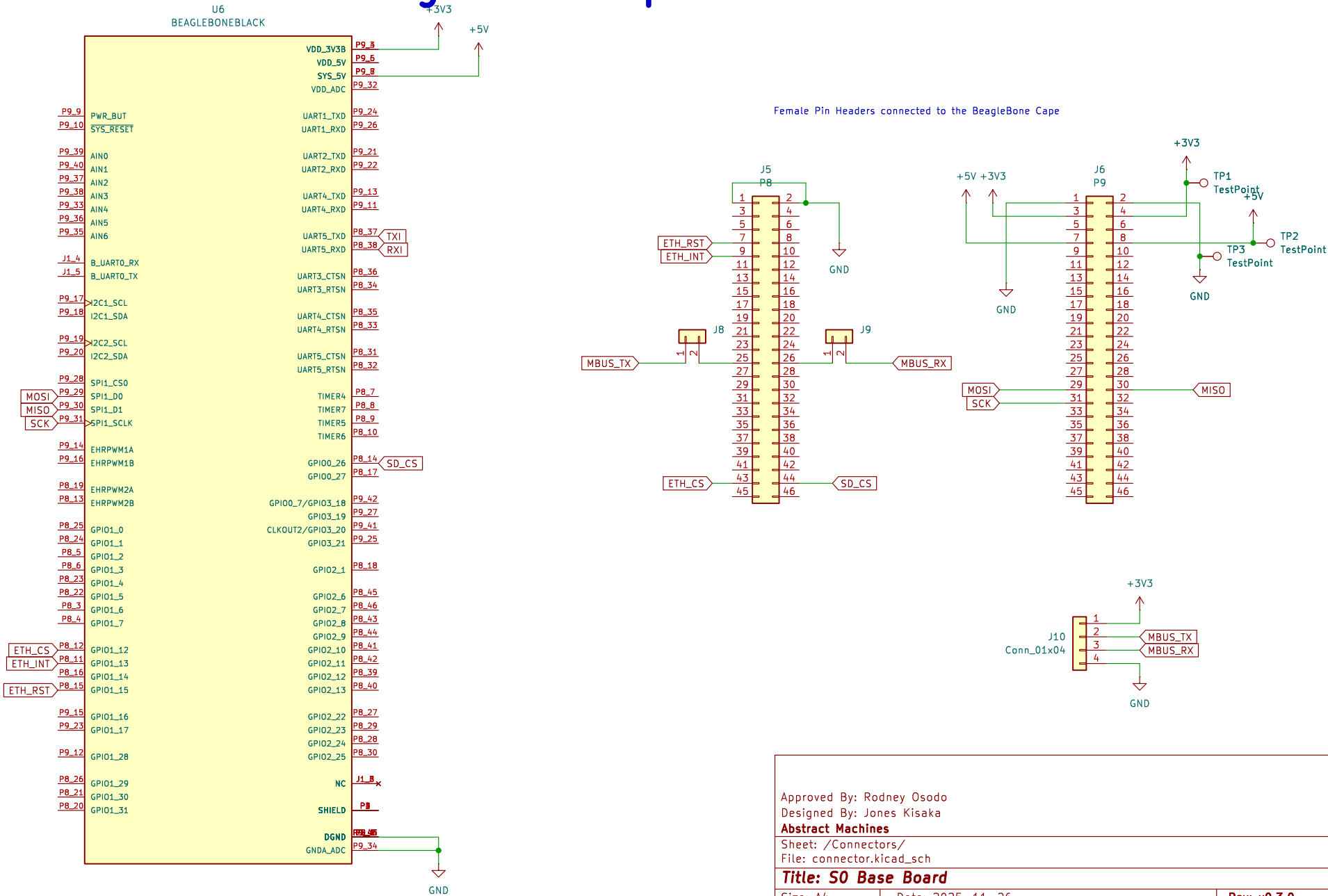
Title: S0 Base Board

Size: A3 Date: 2025-11-26

KiCad E.D.A. 9.0.6

Rev: v0.3.0
Id: 3/6

BeagleBone Cape Board



Designed By: Jones Kisaka

Sheet: /Connectors/
File: connector.kicad_sch

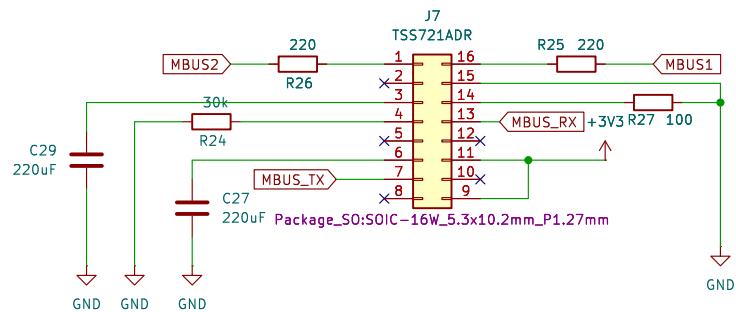
Title: S0 Base Board

Size: A4	Date: 20
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KiCad E.D.A. 9.0.6

4

Id: 4/6



PIN	NAME	DESCRIPTION
1	BUSL2	Meter-Bus
2	VB	Differential Bus Voltage after Rectifier
3	STC	Support Capacitor
4	RIDD	Current Adjust Input
5	PF	Power Fail Output
6	SC	Sampling Capacitor
7	MBUS_TX	Data Output Inverted
8	TX	Data Output
9	BAT	Logic Level Adjust
10	VS	Switch for Bus or Battery Supply output
11	VDD	Voltage Regulator Output
12	RX	Data Input
13	MBUS_RX	Data Input Inverted
14	RIS	Adjust input for modulation current
15	GND	Ground
16	BUSL1	Meter-Bus

Approved By: Rodney Osodo
Designed By: Jones Kisaka

Abstract Machines

Sheet: /wiredmbus/
File: mbus.kicad_sch

Title: S0 Base Board

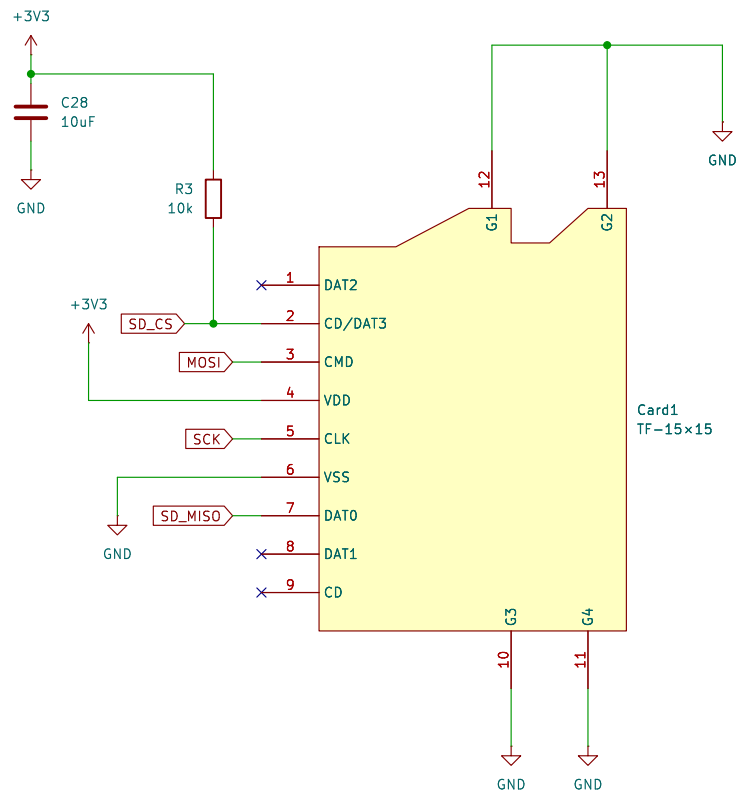
Size: A4 Date: 2025-11-26

KiCad E.D.A. 9.0.6

Rev: v0.3.0

Id: 5/6

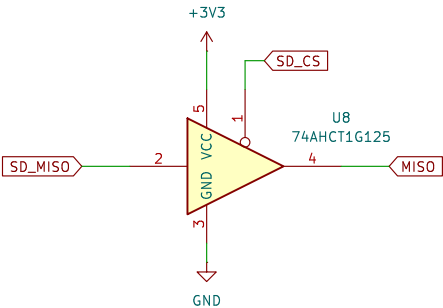
SD CARD



SD Domain to SPI Domain Conversion Table.

PIN	SD DOMAIN	SPI DOMAIN
1	DAT2	NC
2	CD/DAT3	CHIP_SELECT
3	CMD/DATA IN	MOSI
4	VDD	VDD
5	CLK	SERIAL CLOCK
6	VSS	GROUND
7	DAT0/DATA OUT	MISO
8	DAT1	NC
9	CD	NC

SPI MISO ISOLATOR



Approved By: Rodney Osodo

Designed By: Jones Kisaka

Abstract Machines

Sheet: /SDCard/

File: sdcard.kicad_sch

Title: S0 Base Board

Size: A4

Date: 2025-11-26

Rev: v0.3.0

KiCad E.D.A. 9.0.6

Id: 6/6